

SCI-PTC

Correlated-Mode Cleaning and Detector Coefficients

Engineering Conformance Specification

Scientific contract v0.1 – document revision r0.3

TolTEC Scientific Contract Program

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Status. Owner-approved scope; bounded Stage B r0.3 engineering draft. This specification defines conformance evidence but reports no implementation assessment, validation result, or production authorization.

Normative rule. The six shared modules imported below are the sole normative science. This view adds organization and evidence guidance only; it does not introduce an independent equation, estimator, threshold, or scientific interpretation.

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1 Conformance scope and evidence posture

Conformance means that a candidate implementation, its interfaces, and its published products satisfy every applicable shared requirement for the exact realized product role. A source trace or readable file alone is not sufficient. Evidence must bind numerical behavior to immutable identity, declared supports, realized state, response, uncertainty status, and output role.

The present document provides no implementation mapping and no pass/fail disposition. A later authorized conformance effort shall classify each requirement as conformant, nonconformant, partial, unavailable, or not applicable with exact evidence and scope. Unknown is never a pass.

Evidence layer	Required interpretation
Algebraic contract	The shared definitions, equations, assumptions, requirements, and predictions are the authority to be satisfied.
Implementation conformity	Requires source/interface/product evidence for every applicable requirement; unassessed here.
Representation fidelity	Requires proof that serialized state, support, identity, and response represent the realized numerical operation; not performed here.
Observational performance	Requires approved data and thresholds for residual contamination, sky transfer, stability, covariance, and source recovery; not performed here.
Production readiness	Requires separate operational evidence and authorization; not assessed here.

2 Required interface families

An implementation claiming SCI-PTC v0.1 conformance must expose or publish the following scientific interface families. Exact representation is engineering- owned, but every scientific field below must be recoverable without guessing.

Interface family	Required recoverable content
Input identity	Immutable CAL parent; complete RTC parent; observation, scan, segment, sample-time, detector occurrence/UID, array, group, stage, product role, pass, recurrence, and parent identities; exact axes and storage mapping; typed source/residual or surrogate parent where used.
Signal contract	Top-of-atmosphere, point-source-equivalent mJy per fixed nominal beam; factor/atmosphere lineage; validity, uncertainty, nuisance-correlation, and complete upstream-response status.
Support policy	Per-cause decisions for basis fit, loading fit, application, output, coefficient/QC, response, empirical, simulation, and downstream use, including fit-excluded/apply-allowed and the exact inputs required to make each excluded occurrence application-available.
Estimator request	Family, fit objective/loss/metric, missing-data model, candidate set/order/ties, typed conjunctive predicates, centering/scaling, grouping/order, gauge, source protection, refinement, response, covariance, simulation, and outputs.
Learned evidence	Every candidate and fitted state; all predicate values, units, uncertainties, thresholds, comparison polarities, pass/fail causes, ordering/tie result, and selected or rejected disposition.
Resolved model	Exact admitted identities, groups, fitted basis/templates and loadings, modeled subtraction, removed subspace, acting space, rank, degeneracy/gauge, centering/scaling, null space, exact frozen application map and derivative/linear part, classification/refit trace, fallback, and stop state.
Coefficient products	Separate fitted loadings, coordinate parameters, diagnostics, and detector- or detector-time-indexed analysis/gridding families with units, factor domains, normalization operator/domain, support, lifecycle, numerical use, consumers, and prohibited interpretations.

Interface family	Required recoverable content
Response products	PTC-local operator, complete admitted upstream response ending on the CAL grid, complete upstream-to-PTC chain response, realized propagated companions with exact parent domain, full-PTC-procedure signal-response family and typed state differences or explicit unavailability, and optional map-center functional with exact response role and reference-MAP parent.
Uncertainty products	Exact status and conditioning for formal covariance, marginal variance, empirical scatter, calibration/systematic, response, selection, cross-coordinate, and cross-observation terms.
Output/provenance	Transformed TOD role; immutable requested, effective-policy, observation-resolved, learned, resolved, applied, and published state; product-realization and role-availability axes; required siblings, completeness, failure scope, random/surrogate state, and typed relations to downstream consumers.

3 Fail-closed implementation behavior

Failure is scoped to the affected product or claim unless a required product dependency makes the entire requested output incomplete. Implementations shall represent missing, non-finite, invalid, rejected, disabled, not requested, unavailable, and numeric zero as distinct states. Product realization, response availability, covariance availability, and evidence disposition are independent axes. Implementations shall not infer missing units, identities, supports, thresholds, rank rules, response terms, bias claims, or covariance.

A typed fallback is conformant only when the request authorizes it and the realized state records its cause, stage, changed scientific operation, response consequence, covariance consequence, and affected availability. Required-output failure propagates to the calling boundary. Best-effort diagnostic failure is inert only when that diagnostic is neither required nor used by policy.

The base-v0.1 conditioned- r branch is diagnostic-only. A conformance trace must show that it cannot change calibrated- x fit membership, subtraction, output, or coefficients. Raw- r processing, r -derived subtraction from x , joint x/r PCA, raw Beammap PTC, cross-array fitting, and polarimetry are outside this contract.

4 Evidence checklist for a later assessment

For each claimed product role, a later assessment should obtain:

1. the immutable requested, effective-policy, observation-resolved, learned/fitted, resolved-selected, applied/realized, and published states and their parent links;
2. a field-level source/interface trace for every shared requirement;
3. minimal deterministic probes for the applicable shared predictions;
4. exact comparisons of fitted support, candidate evidence, removed subspace, frozen application map, classification, response-domain chain, covariance status, expectation/bias status, and coefficient semantics;
5. proof that PCA/SVD application projects through the declared frozen subspace/metric in the declared acting space, that fixed-state companions cannot affect learning, that the complete upstream response ending on the CAL grid is applied exactly once, and that full-procedure response restarts from the immutable CAL parent while retaining typed state changes;
6. proof that every fit-excluded/apply-allowed occurrence has the exact basis-coefficient or recomputation input, neighboring support, detector binding, metric, coordinate, and boundary state required by the resolved operator, and otherwise becomes unavailable without inferred interpolation, zero filling, reconstruction, or extension;
7. failure probes for missing external-parent identity, mismatched surrogate membership, zero/empty support, non-finite values, rank deficiency, oscillation, unavailable response, independent realization/availability states, and incomplete publication; and
8. an evidence-layer disposition that does not promote conformance into observational performance or production authorization.

The shared prediction register is deliberately implementation-independent. A later probe may falsify a conformance claim, but passing a small synthetic case does not by itself validate observational performance. The science-team validation table groups the intended studies and observables; it supplies no threshold, result, or evidence-layer promotion.

5 Normative notation

Symbol	Meaning
o, s, g	Immutable observation, scan, and coherent-segment identities.
t, d	Sample position on the declared time grid and immutable detector occurrence/UID.
a, n, ℓ	Array, readout network/electronics group, and optional local or focal-plane group.
q	Immutable product-stage identity; it is never a diagnostic statistic.
$Y \equiv Y^{\text{CAL}}$	Admitted sample-by-detector calibrated x matrix, in top-of-atmosphere point-source-equivalent mJy per fixed nominal beam.
S	Astronomical contribution on the admitted detector-time grid.
$M_{\star} = [m_{\star, tk}]$	Latent shared temporal-mode or supplied-template representation admitted by the signal model.
$A_{\star} = [a_{\star, dk}]$	Latent detector-loading representation for the admitted correlated component.
$U_{\star} = M_{\star} A_{\star}^{\top}$	Correlated-mode or supplied-template component admitted as the estimand; it is not yet a fitted estimate.
$\hat{U} = \hat{M} \hat{A}^{\top}$	Realized fitted estimate of the correlated component in the calibrated signal unit.
$Z \equiv Y^{\text{PTC}}$	Transformed calibrated detector timestream.
N	Remaining detector-noise term; it is not assumed independent or stationary.
$\mathcal{G}$	Declared within-array hierarchy and its joint or sequential fit order.
C, D, λ	Centering operator, invertible internal scaling operator, and declared centering-location state.
$\hat{U}$	Realized removed subspace, represented independently of basis gauge.
$\mathcal{N}_{\text{PTC}}$	Declared PTC null space, including additive-reference losses.
$\mathcal{S}^j$	Support for use j : basis fit, loading fit, application, output, coefficient/QC, response, empirical, simulation, or downstream use.
$\Phi, \mathcal{I}$	Typed direct causes and transitive causal-influence record.
R, E, O	Requested, effective-policy, and observation-resolved input states.
$\mathcal{L}$	Complete learned/fitted evidence for every evaluated candidate, before final selection.
Θ	Immutable resolved selected model, including the exact application map and all selection evidence.
Λ, Π	Applied/realized state and published product state.
$\mathcal{A}_{\Theta}$	Exact frozen-state numerical application map acting on an admitted detector-time input.
$J_{\Theta}[Y]$	Fréchet derivative $D\mathcal{A}_{\Theta}[Y]$, or the declared finite-dimensional linear part when $\mathcal{A}_{\Theta}$ is affine.
$\mathcal{L}_{\Theta}^t, \mathcal{L}_{\Theta}^d, \mathcal{L}_{\Theta}^v$	Frozen projection operators in temporal, detector, or explicitly vectorized detector-time space.
$\mathcal{P} = (\mathcal{P}_Z, \mathcal{P}_{\Lambda})$	Full data-dependent PTC procedure and its transformed-signal and typed-state projections.
$K_{\text{up} \rightarrow \text{CAL}}$	Complete admitted upstream response carried by the CAL product from a declared source-domain perturbation to the CAL detector-time grid, including every applicable admitted source-to-detector, beam-convention, coordinate/scan, RTC, and CAL operation; otherwise an explicit unavailable state.
τ_{src}	Exact named source-domain perturbation/template.
$k_{\text{src}}^{\text{CAL}}$	Realized source companion already on the admitted CAL detector-time grid; when source-domain propagation is requested, $k_{\text{src}}^{\text{CAL}} = K_{\text{up} \rightarrow \text{CAL}} \tau_{\text{src}}$.
$k_{\text{src}}^{\text{PTC} \Theta}$	Fixed-state propagated detector-time companion $J_{\Theta}[Y] k_{\text{src}}^{\text{CAL}}$.
$K_{\text{PTC} \Theta}$	Complete fixed-state chain response $J_{\Theta}[Y] \circ K_{\text{up} \rightarrow \text{CAL}}$.
$K_{\text{PTC}}^{\text{proc}}$	Full-PTC-procedure finite-perturbation signal-response family from the immutable CAL parent.

Symbol	Meaning
Δ_{state}	Typed comparison of two discrete or mixed realized states; it is not numerical subtraction.
A_{ref}	Exact named reference MAP functional, imported only for an optional diagnostic.
ρ_{center}	Optional estimated map-center point-source response diagnostic.
C_Y, C_Z	Declared input and transformed-signal covariance, with exact status and conditioning.
η_i	Declared diagnostic statistic for coefficient index i , where $i = d$ or $i = (t, d)$ according to the coefficient family.
γ_i	A downstream analysis/gridding coefficient with declared detector or detector-time index family; never a fitted loading by implication.
Avail_r	Typed availability for requested role r : <code>computed_published</code> , <code>not_computed_or_not_requested_for_this_product</code> , <code>invalid</code> , or <code>unavailable</code> .
Exist_p	Product realization state for product p : <code>realized</code> , <code>disabled</code> , <code>not_requested</code> , <code>rejected</code> , or <code>unavailable</code> .

Superscripts and subscripts identify scientific role and conditioning, not a storage layout. Dense row or column positions are local conveniences and are never stable external identities. Operator products are written only when their domains and codomains are compatible.

6 Normative definitions

SCI-PTC-DEF-001 – Admitted CAL parent. The immutable SCI-CAL detector-time product accepted by all required unit, identity, validity, lineage, uncertainty, and response gates. It is the numerical parent of every base-v0.1 fit and support-changing refit.

SCI-PTC-DEF-002 – Correlated-mode estimand. A latent declared detector-time component $U_\star$ represented by a shared temporal basis or supplied template and detector loadings. PTC estimates $\hat{U}$ and the realized removed subspace $\hat{\mathcal{U}}$; statistical correlation does not identify their physical origin.

SCI-PTC-DEF-003 – Transformed detector signal. The calibrated detector-time signal after subtracting the realized correlated component and restoring every invertible internal coordinate transform. It is not an independent sky estimate.

SCI-PTC-DEF-004 – Removed subspace. The complete realized subspace or template span whose fitted contribution is subtracted, including its group, support, metric, rank, order, and gauge-equivalence class.

SCI-PTC-DEF-005 – Null space. The set of admitted input perturbations that the complete realized operation maps to zero or makes unidentifiable, including additive-reference loss and any sky component overlapping the removed subspace.

SCI-PTC-DEF-006 – Gauge. A sign, scale, ordering, or rotation freedom in a basis/loading representation that leaves the fitted correlated component and removed subspace unchanged.

SCI-PTC-DEF-007 – Centering state. The declared location estimate, axis, population, weights, support, masks, boundary, unit, and restoration rule used to define the fitting coordinate.

SCI-PTC-DEF-008 – Scaling state. The declared nonzero internal scale, axis, population, weights, estimator, support, unit, and inverse transform used for numerical conditioning.

SCI-PTC-DEF-009 – Basis-fit support. Detector-time occurrences permitted to influence construction or selection of a temporal basis or supplied-template fit.

SCI-PTC-DEF-010 – Loading-fit support. Occurrences permitted to influence the detector loading of an already declared temporal basis or template.

SCI-PTC-DEF-011 – Application support. Occurrences to which the exact frozen application map is authorized to apply, including an explicitly allowed fit-excluded/apply-allowed state only where the resolved operator defines the value from admitted frozen state and support.

- SCI-PTC-DEF-012 – Output support.** Occurrences eligible to remain in a transformed output; it is independent of fit and application supports.
- SCI-PTC-DEF-013 – Cause-to-support policy.** An immutable mapping from each typed cause to separate basis-fit, loading-fit, application, output, coefficient/QC, response, empirical, simulation, and downstream-support decisions.
- SCI-PTC-DEF-014 – Fixed-state application map.** The exact numerical map $\mathcal{A}_\Theta$ obtained after all fitted, selected, classified, ranked, grouped, masked, metric, support, coordinate, and fallback state is frozen. For an admitted PCA/SVD family it projects any admitted application input through the frozen realized subspace and metric, with its acting space and coefficient recomputation declared.
- SCI-PTC-DEF-015 – Full PTC procedure.** The complete learn, fit, select, rank, classify, refit, apply, and publish operation beginning from the immutable admitted CAL parent.
- SCI-PTC-DEF-016 – Response companion.** An exact perturbation with a declared domain and parent, carried through the exact fixed-state derivative or used in a full-procedure injection study. A detector-time companion already on the CAL grid is not passed through $K_{\text{up} \rightarrow \text{CAL}}$ again, and no companion silently enters the fit.
- SCI-PTC-DEF-017 – Conditional response.** The derivative or declared affine linear part of the frozen application map, conditioned on the exact resolved selected model. It excludes movement of learned, selected, classified, or fallback state.
- SCI-PTC-DEF-018 – Full-procedure response.** A finite-perturbation signal response produced by rerunning the complete PTC procedure from the immutable admitted CAL parent. It is paired with a typed comparison of changed discrete/mixed state and may be a family over amplitude, side, source state, or location.
- SCI-PTC-DEF-019 – Post-fit detector assessment.** A bounded evaluation of declared residual, loading, influence, stability, source-response, and diagnostic quantities used to classify detector state.
- SCI-PTC-DEF-020 – Support-changing refinement.** A post-fit classification change that alters basis-fit or loading-fit support and therefore requires one complete refit from the immutable admitted CAL parent.
- SCI-PTC-DEF-021 – PTC pass.** A complete PTC realization with an immutable input parent and distinct realized identity. It is neither an internal solver iteration nor an external FRUIT recurrence.
- SCI-PTC-DEF-022 – Diagnostic-only conditioned r .** A separately authorized, PTC-compatible, electronics-enriched r product used only for inert or advisory diagnostics in base v0.1. It cannot control or subtract calibrated x .
- SCI-PTC-DEF-023 – Fitted loading.** A gauge-dependent coefficient coupling a temporal basis/template to one detector. It is not a gridding weight, precision, significance, or sensitivity.
- SCI-PTC-DEF-024 – Analysis/gridding coefficient.** An explicitly named detector-indexed or detector-time-indexed coefficient permitted for downstream aggregation. Its index family, identity, unit, normalization operator and domain, factors, support, and lifecycle are declared; it is nonprecision unless separately proven.
- SCI-PTC-DEF-025 – Typed availability.** A role-specific response, covariance, coefficient, or diagnostic status among `computed_published`, `not_computed_or_not_requested_for_this_product`, `invalid`, or `unavailable`, with a cause and affected claim. It is distinct from whether a product exists or was realized.
- SCI-PTC-DEF-026 – Least-aggressive admissible candidate.** The first candidate under a declared ordering for which every required residual-contamination, astronomical-transfer, conditioning, support, stability, and QC predicate passes. No failed predicate can be offset by a scalar score.
- SCI-PTC-DEF-027 – Map-center response diagnostic.** The optional scalar functional of an exact source companion, propagated PTC response, exact named reference MAP operator, and center selector. It is neither the general PTC response nor MAP authority.
- SCI-PTC-DEF-028 – Material state.** Every requested, resolved, learned, fitted, selected, classified, applied, randomized, fallback, or publication quantity capable of changing a declared output, response, coefficient, uncertainty, or downstream use.
- SCI-PTC-DEF-029 – Learned/fitted evidence state.** The complete record $\mathcal{L}$ of every evaluated candidate, its fitted state, diagnostics, predicate values and units, pass/fail causes, ordering, tie result, and disposition before final selection.
- SCI-PTC-DEF-030 – Resolved selected model.** The immutable state Θ selected from learned evidence, including its exact basis/subspace, acting space, metric, coefficient-recomputation rule, supports, coordinates, grouping, rank, classes, fallback, and application map.

SCI-PTC-DEF-031 – Frozen-subspace projection family. An admitted PCA/SVD application family that recomputes application coefficients linearly against a frozen realized subspace and metric. It must declare temporal-left, detector-right, two-sided, detector/time-specific, or general vectorized action.

SCI-PTC-DEF-032 – Frozen-component subtraction family. A separately named affine family $Y' \mapsto Y' - \hat{U}$ with one frozen numerical component. Its derivative is the identity; it is not interchangeable with frozen-subspace projection and is not the base-v0.1 PCA/SVD application rule.

SCI-PTC-DEF-033 – Realized propagated companion. The detector-time companion after exactly one compatible application of the requested local or chain response. It records whether its parent was source-domain or already on the CAL grid and whether propagation was conditional or full-procedure.

SCI-PTC-DEF-034 – Typed state comparison. A field-aware comparison Δ_{state} of learned, selected, classified, ranked, grouped, support, convergence, and fallback state across two procedure runs. Discrete state is compared by identity and disposition, never numerically subtracted.

SCI-PTC-DEF-035 – Product realization state. A state among `realized`, `disabled`, `not_requested`, `rejected`, or `unavailable` for a named product role. It is orthogonal to response or uncertainty availability.

SCI-PTC-DEF-036 – External source/residual parent. A source model, mask, residual, or prior-pass parent admitted only with explicit owner, unit, coordinate frame, registration, validity, response, support, and recurrence identity.

SCI-PTC-DEF-037 – Shifted/null surrogate. A derived realization in which signal, associated validity/eligibility, support, boundaries, randomness, and parent identity are transformed together under one declared shift or null rule.

SCI-PTC-DEF-038 – Astronomical-transfer predicate. A candidate-admission predicate bound to an exact source class and companion, conditional or full-procedure response type, amplitude/location domain, metric, tolerance, mask/support, upstream response identity, and response-uncertainty treatment.

SCI-PTC-DEF-039 – Expectation/bias claim. A family- and role-specific statement about the conditional expectation or bias of astronomical signal, fitted nuisance, remaining detector noise, calibration uncertainty, and selection. A missing or unproved claim is explicitly unavailable rather than silently unbiased.

SCI-PTC-DEF-040 – Application-available occurrence. An occurrence, including a fit-excluded occurrence, for which the resolved frozen operator has every required detector binding, basis coefficient or coefficient-recomputation input, neighboring support, metric term, coordinate transform, and boundary state needed to define the application value without interpolation, zero filling, or an inferred extension.

SCI-PTC-DEF-041 – Complete upstream response ending on the CAL grid. The exact admitted response $K_{\text{up} \rightarrow \text{CAL}}$ carried by the CAL product from a declared source-domain perturbation to the CAL detector-time grid. It composes every applicable admitted source-to-detector relation, beam convention, coordinate and scan geometry, RTC operation, and CAL operation; omission or unavailability of a required factor makes the complete-response claim unavailable.

7 Normative equations

The organizing detector-time model separates the latent correlated component from its fitted estimate:

$$Y_{td}^{\text{CAL}} = S_{td} + U_{*,td} + N_{td}, \quad U_{*,td} = \sum_{k=1}^K M_{*,tk} A_{*,dk}. \quad (1)$$

For the immutable fitted parent, the realized modeled component and transformed signal satisfy the identity

$$\hat{U} = \hat{M} \hat{A}^T, \quad Z = \mathcal{A}_{\Theta}(Y^{\text{CAL}}) = Y^{\text{CAL}} - \hat{U}. \quad (2)$$

Equation 2 states the realized result on the fitted parent. It does not define how a new input or response companion is applied.

For a declared centering operator C with location state λ and an invertible internal scaling D , the fitting coordinate and physical-unit application map are

$$\tilde{Y}' = D^{-1} C(Y'; \lambda), \quad \mathcal{A}_{\Theta}(Y') = C^{-1} \left(D \tilde{\mathcal{A}}_{\Theta}(\tilde{Y}'); \lambda \right). \quad (3)$$

Every reversible coordinate is restored for ordinary output. Any irrecoverable additive mode is published in $\mathcal{N}_{\text{PTC}}$.

Every typed cause ϕ is mapped independently to each use:

$$\Gamma(\phi) = (\Gamma_{\text{basis}}, \Gamma_{\text{load}}, \Gamma_{\text{apply}}, \Gamma_{\text{out}}, \Gamma_{\text{coeff}}, \Gamma_{\text{resp}}, \Gamma_{\text{emp}}, \Gamma_{\text{sim}}, \Gamma_{\text{down}}). \quad (4)$$

An admitted weighted or robust low-rank family minimizes

$$(\widehat{M}, \widehat{A}) = \arg \min_{M, A} \sum_{t, d} w_{td}^{\text{fit}} \rho(\tilde{Y}_{td} - (MA^T)_{td}), \quad (5)$$

where $w_{td}^{\text{fit}} = 0$ denotes no fit influence, not a zero-valued observation. The loss, metric, support, rank, and missing-data semantics are declared.

For a fixed supplied design B and fit metric Ω , a conditioned regression family is

$$\widehat{c} = (B^T \Omega B)^+ B^T \Omega y, \quad \mathcal{L}_{B, \Omega} = I - B(B^T \Omega B)^+ B^T \Omega. \quad (6)$$

The generalized inverse, tolerance, units, gauge, coefficient support, and application support are material state.

A weighted common-mode family may estimate its temporal mode as

$$\widehat{m}_t = \frac{\sum_d w_{td}^{\text{fit}} \omega_{td}^{\text{cm}} \tilde{Y}_{td}}{\sum_d w_{td}^{\text{fit}} \omega_{td}^{\text{cm}}}, \quad (7)$$

only where the denominator is finite and nonzero. The aggregation weight ω_{td}^{cm} , its unit, normalization, and population are declared; it is not a fitted detector loading by implication.

For base-v0.1 PCA/SVD, application means projection of any admitted application input through the frozen realized subspace and metric:

$$\tilde{\mathcal{A}}_{\Theta}(\tilde{Y}') = \begin{cases} \mathcal{L}_{\Theta}^t \tilde{Y}', & \text{temporal-space left action,} \\ \tilde{Y}' \mathcal{L}_{\Theta}^d, & \text{detector-space right action,} \\ \text{unvec}[\mathcal{L}_{\Theta}^v \text{vec}(\tilde{Y}')], & \text{declared vectorized action.} \end{cases} \quad (8)$$

The realized family selects exactly one compatible action and records its domain, codomain, metric, supports, generalized inverse/tolerance, and linear coefficient-recomputation rule. A two-sided or detector/time-specific action is represented by its exact vectorized operator; it is not inferred from matrix orientation or rank alone.

A separately named frozen-component subtraction family is

$$\mathcal{A}_{\widehat{U}}^{\text{sub}}(Y') = Y' - \widehat{U}, \quad D\mathcal{A}_{\widehat{U}}^{\text{sub}}[Y](H) = H. \quad (9)$$

It is not interchangeable with Equation 8 and is not the base-v0.1 PCA/SVD application rule.

A declared within-array hierarchical model may be written

$$U_{\star} = U_{\star}^{\text{array}} + U_{\star}^{\text{network}} + U_{\star}^{\text{local}}, \quad (10)$$

with exact group identities and joint or sequential fit order. Sequential fitting publishes the complete ordered composition and cumulative response.

Let $\mathcal{C}$ be a finite candidate set with declared order $\preceq$. The base selection rule is

$$c_{\star} = \min_{\preceq} \{c \in \mathcal{C} : P_{\text{corr}}(c) \wedge P_{\text{sky}}(c) \wedge P_{\text{cond}}(c) \wedge P_{\text{supp}}(c) \wedge P_{\text{stab}}(c) \wedge P_{\text{QC}}(c)\}. \quad (11)$$

Every predicate value, unit, threshold, uncertainty treatment, pass/fail cause, order, tie, and rejected candidate is retained in $\mathfrak{L}$.

For refinement index j , the support-changing lifecycle is

$$\begin{aligned} \Theta_j &= \text{Fit}(Y^{\text{CAL}}; \mathcal{S}_j), \\ \mathcal{S}_{j+1} &= \text{Classify}(Y^{\text{CAL}}, \Theta_j), \\ \Theta_{j+1} &= \text{Fit}(Y^{\text{CAL}}; \mathcal{S}_{j+1}). \end{aligned} \quad (12)$$

Every fit uses the same immutable admitted CAL parent. A cleaned Z_j is not the next fit parent, and the final complete model is applied once.

The one-way state lifecycle is

$$R \longrightarrow E \longrightarrow O \longrightarrow \mathcal{L} \longrightarrow \Theta \longrightarrow \Lambda \longrightarrow \Pi, \quad (13)$$

where learned evidence, resolved selection, numerical application, and publication are distinct immutable states.

The full PTC procedure has separately typed signal and state projections:

$$\mathcal{P}(Y^{\text{CAL}}, R, E, O) = (\mathcal{P}_Z(Y^{\text{CAL}}), \mathcal{P}_\Lambda(Y^{\text{CAL}})) = (Z, \Lambda). \quad (14)$$

For fixed resolved model Θ , the local detector-time response and a realized CAL-grid companion are

$$J_\Theta[Y] = D\mathcal{A}_\Theta[Y], \quad k_{\text{src}}^{\text{PTC}|\Theta} = J_\Theta[Y] k_{\text{src}}^{\text{CAL}}. \quad (15)$$

If the parent is source-domain, the complete chain response is

$$k_{\text{src}}^{\text{CAL}} = K_{\text{up} \rightarrow \text{CAL}} \tau_{\text{src}}, \quad K_{\text{PTC}|\Theta} = J_\Theta[Y] \circ K_{\text{up} \rightarrow \text{CAL}}, \quad k_{\text{src}}^{\text{PTC}|\Theta} = K_{\text{PTC}|\Theta} \tau_{\text{src}}. \quad (16)$$

Equations 15–16 are alternative entry points for different companion parents; $K_{\text{up} \rightarrow \text{CAL}}$ is never applied twice.

For a declared detector-time perturbation h^{CAL} , the full-procedure finite-perturbation signal response is

$$K_{\text{PTC}}^{\text{proc}}(\epsilon, h^{\text{CAL}}) = \frac{\mathcal{P}_Z(Y^{\text{CAL}} + \epsilon h^{\text{CAL}}) - \mathcal{P}_Z(Y^{\text{CAL}})}{\epsilon}. \quad (17)$$

The corresponding state evidence is retained separately:

$$\Delta_{\text{state}}(\epsilon, h^{\text{CAL}}) = \text{Compare}_{\text{typed}}[\mathcal{P}_\Lambda(Y^{\text{CAL}} + \epsilon h^{\text{CAL}}), \mathcal{P}_\Lambda(Y^{\text{CAL}})]. \quad (18)$$

Amplitude, side, source state, location, domain, and every changed rank, mask, group, support, class, convergence, or fallback field are part of the result.

For response role $r \in \{\text{fixed}, \text{procedure}\}$, the optional map-center diagnostic is

$$\rho_{\text{center}}^r = e_{\text{center}}^T A_{\text{ref}} k_{\text{src}}^{\text{PTC}, r}. \quad (19)$$

The exact response role, source-companion parent, reference MAP operator, and center selector are mandatory factors. The diagnostic confers no MAP or general beam authority.

For a fixed differentiable realized map, conditional covariance is

$$C_{Z|\Theta} = J_\Theta[Y] C_Y J_\Theta[Y]^T. \quad (20)$$

For random selected state Θ , the exact decomposition is

$$\text{Cov}(Z) = \text{E}[\text{Cov}(Z | \Theta)] + \text{Cov}(\text{E}[Z | \Theta]). \quad (21)$$

Omitting the second term is a conditioned approximation, not complete uncertainty.

A diagnostic indexed by detector or detector-time occurrence may take the generic residual form

$$\eta_i = H_i(Z, \hat{U}, \mathcal{S}, \Theta; \chi_H), \quad (22)$$

where χ_H records the domain, population/model reference, normalization, support, uncertainty, and policy role.

A downstream analysis/gridding coefficient family is

$$\tilde{\gamma}_i = f_\gamma(\eta_i; \chi_\gamma), \quad \gamma_i = \mathcal{N}_\gamma(\tilde{\gamma}_i; \{\tilde{\gamma}_j : j \in \mathcal{D}_\gamma\}), \quad (23)$$

where $i = d$ or $i = (t, d)$ is declared, $\mathcal{D}_\gamma$ is the exact normalization domain, and f_γ and $\mathcal{N}_\gamma$ are named operators. The result is not precision unless separately proved.

Transitive causal influence for output j is

$$\mathcal{I}_j = \bigcup_{i: i \rightarrow j} (\{i\} \cup \mathcal{I}_i), \quad (24)$$

including inputs that affect fitted state or selection even when a local numeric derivative vanishes.

For disabled PTC on the PTC-dependent route,

$$\text{PTC} = \text{disabled} \implies \text{Exist}_p = \text{disabled} \quad \text{for } p \in \{Z, \Lambda, \gamma, \text{MAP}_{\text{PTC-route}}\}. \quad (25)$$

This product-realization statement does not assign response availability and does not govern a separately authorized direct CAL-to-MAP route.

8 Normative assumptions and declared limits

- SCI-PTC-ASM-001.** The primary numerical parent is an admitted SCI-CAL calibrated x product in the declared top-of-atmosphere, point-source-equivalent mJy-per-fixed-nominal-beam convention.
- SCI-PTC-ASM-002.** The complete RTC and CAL lineage, including exact detector/time identities, validity, influence, response, and uncertainty status, is available or the affected claim fails closed.
- SCI-PTC-ASM-003.** Correlation alone does not identify atmosphere, electronics, calibration residual, detector noise, astronomical signal, or any unique mixture.
- SCI-PTC-ASM-004.** PTC fitting is confined to one array unless a successor authority supplies a cross-array spectral, calibration, beam, source-color, and response model.
- SCI-PTC-ASM-005.** The selected estimator family supplies explicit missing-data semantics. Replacing a missing or excluded value with numerical zero is not exclusion unless the estimator proves equivalence.
- SCI-PTC-ASM-006.** Every centering/scaling transform has a declared axis, population, support, weights, estimator, boundary, unit, inverse, gauge, and null-space consequence.
- SCI-PTC-ASM-007.** Internal scaling is invertible and restored before ordinary transformed output; any noninvertible operation changes the signal definition and requires a separately named output role.
- SCI-PTC-ASM-008.** A source mask protects only its declared model and registered support. Unmasked compact or extended sky components may overlap the fitted subspace.
- SCI-PTC-ASM-009.** Candidate selection is finite, conjunctive, deterministically ordered, and least-aggressive. Numerical predicates are owner-controlled inputs rather than inferred defaults.
- SCI-PTC-ASM-010.** All support-changing detector refinements refit the complete selected model from the same immutable admitted CAL parent.
- SCI-PTC-ASM-011.** A sequential residual fit is valid only when declared as part of one complete ordered hierarchical estimator with cumulative removed subspace, response, covariance, and parentage.
- SCI-PTC-ASM-012.** The conditioned- r branch, when available, has its own operator, unit, response, validity, optical-leakage state, and exact relation to the calibrated- x grid.
- SCI-PTC-ASM-013.** Base-v0.1 r analysis is inert or advisory. It cannot change calibrated- x fit membership, subtraction, output retention, or coefficients and cannot supply an x subtraction basis.
- SCI-PTC-ASM-014.** A fixed-state application map does not by itself represent the response of data-derived rank, mask, grouping, clipping, classification, convergence, or fallback decisions.
- SCI-PTC-ASM-015.** A whole-chain RTC-to-CAL-to-PTC injection is distinct from the PTC full-procedure response and requires separately named upstream owners and response companions.
- SCI-PTC-ASM-016.** Full covariance is optional. An absent covariance or cross-covariance is unknown/unavailable rather than zero, independent, or diagonal.
- SCI-PTC-ASM-017.** Fitted loadings, centering/scaling parameters, diagnostic quantities, and analysis/gridding coefficients are distinct types and lifecycles.
- SCI-PTC-ASM-018.** A scalar analysis/gridding coefficient is nonprecision unless its exact target variance, conditioning, unit, normalization, covariance, and independence conditions are proven.
- SCI-PTC-ASM-019.** PTC internal iteration, a new PTC pass, and an external FRUIT recurrence are distinct identities with distinct parents and stop rules.
- SCI-PTC-ASM-020.** The in-memory transformed TOD is an authoritative intermediate for the PTC-dependent route, not an independent sky estimator or a claim of map, beam, significance, or production validity.
- SCI-PTC-ASM-021.** A required unavailable input, response term, support, identity, or output makes only the affected product or claim unavailable; it is never replaced silently by a numeric default.
- SCI-PTC-ASM-022.** Implementation conformity, representation fidelity, observational performance, and production readiness require separate evidence and are not established by this contract.
- SCI-PTC-ASM-023.** For every admitted base-v0.1 PCA/SVD family, application to science or a fixed-state companion is projection through the frozen realized subspace and metric with linear coefficient recomputation. Subtraction of one frozen fitted numerical component is a distinct family and is not inferred.
- SCI-PTC-ASM-024.** No universal temporal, detector, two-sided, or vectorized PCA action is inferred from rank or matrix orientation. Each realized family declares exactly which space contains the removed basis and the exact left, right, or general action.

SCI-PTC-ASM-025. Source-domain templates, CAL-grid detector-time companions, PTC-local operators, complete chain responses, and realized propagated companions are distinct typed objects. An upstream response operator is applied exactly once.

SCI-PTC-ASM-026. Learned candidate evidence, resolved selected model, applied state, and published product state are distinct immutable lifecycle stages even when one implementation stores them together.

SCI-PTC-ASM-027. An external source/residual parent or shifted/null surrogate is scientifically usable only when its complete parent, coordinate, validity, support, response, recurrence, and transformation state is available; insufficient support is not a valid zero surrogate.

SCI-PTC-ASM-028. Fit exclusion does not itself authorize application. A fit-excluded occurrence is application-available only when the resolved frozen operator defines that occurrence from admitted frozen state and support without inferred interpolation, zero filling, or extension.

9 Normative requirements

9.1 Input identity, lineage, and state

SCI-PTC-REQ-001 – Admitted calibrated signal. PTC shall accept as its primary numerical parent only an admitted sample-by-detector SCI-CAL x product in top-of-atmosphere, point-source-equivalent mJy per fixed nominal beam, with explicit response state.

SCI-PTC-REQ-002 – Complete RTC lineage. The admitted bundle shall retain conditioned- x identity, raw- r parent identity, exact time grid, and every causal RTC selector, segmentation, validity, response, uncertainty, replacement, and influence state.

SCI-PTC-REQ-003 – Upstream availability. An invalid or unavailable required RTC or CAL fact shall make the affected calibrated PTC product or claim unavailable; PTC shall not repair, approximate, or relabel it.

SCI-PTC-REQ-004 – Matrix shape. Every primary matrix shall declare samples on rows, detectors on columns, actual extents, flattening/storage order, and the bijection between positions and scientific identities.

SCI-PTC-REQ-005 – Time identity. Sample time shall carry declared scale, epoch, unit, actual timestamps, gaps, segment boundaries, and any cross-boundary support; nominal rate shall not replace actual time identity.

SCI-PTC-REQ-006 – Detector identity. Detector joins shall use immutable occurrence/UID identity and exact APT binding; dense columns, file order, or local integers shall not be external identities.

SCI-PTC-REQ-007 – Within-array domain. Every base fit shall be confined to one declared array. Cross-array modes shall be unavailable without separately authorized spectral, calibration, beam, source-color, and response authority.

SCI-PTC-REQ-008 – Requested operation. The requested state shall identify estimator family, candidate ordering, conjunctive predicates, centering/scaling, gauge, grouping, support policy, refinement, source protection, coefficient families, response, covariance, simulation, and output roles; absence shall differ from an explicit value.

SCI-PTC-REQ-009 – One-way state lifecycle. Requested, effective-policy, observation-resolved input, learned/fitted evidence, resolved selected model, applied/realized state, and published product state shall be immutable, one-way linked, and distinct as in Equation 13. A later stage shall not overwrite or reconstruct an earlier stage.

SCI-PTC-REQ-010 – Unit preservation limit. Ordinary transformed x shall retain the admitted CAL unit and calibration convention, but unit retention shall not be labeled preservation of point-source peak, absolute level, extended-source response, detector combinations, or beam shape.

9.2 Validity, causes, and support

SCI-PTC-REQ-011 – Cause is not action. Every flag shall remain a typed cause. No cause name alone shall imply zero-fill, universal rejection, zero coefficient, refit, or downstream ineligibility.

SCI-PTC-REQ-012 – Per-use support map. Each cause shall map independently and immutably into basis-fit, loading-fit, application, output, coefficient/QC, response, empirical, simulation, and downstream supports as in Equation 4.

SCI-PTC-REQ-013 – Finite eligible arithmetic. Only eligible finite occurrences shall enter a fit, normalization, covariance estimate, convergence statistic, or coefficient statistic. Missing, non-finite, invalid, rejected, unavailable, and numeric zero shall remain distinct.

SCI-PTC-REQ-014 – No silent zero filling. A missing or fit-excluded value shall not be replaced by numerical zero before PCA/SVD or another estimator unless exact equivalence to zero statistical influence is proved for the selected estimator.

SCI-PTC-REQ-015 – Fit-excluded apply-allowed state. The support model shall express an occurrence excluded from mode or loading learning but eligible for application and output, without treating that state as fit validity.

SCI-PTC-REQ-016 – Direct synthesized/replaced occurrences. Direct ALIGN-synthesized or RTC-replaced occurrences shall be excluded from the calibrated PTC science branch.

SCI-PTC-REQ-017 – Transitive influence. Noncenter causal influence shall be preserved and evaluated under PTC's declared use-specific support and response policy; compact over-approximation is permitted and under-approximation is not.

SCI-PTC-REQ-018 – Mask identity. Every mask shall declare polarity, use, parent, frame, unit, registration, support, boundary convention, and derivation state. One mask bit shall not silently govern every use.

9.3 Estimand, coordinates, and grouping

SCI-PTC-REQ-019 – Declared estimand. Every realized operation shall identify the exact correlated-mode or supplied-template estimand, objective/loss, metric, parameterization, fit support, loading support, application support, acting space, and frozen-state application map.

SCI-PTC-REQ-020 – No physical-origin inference. A fitted correlated component shall not be labeled atmosphere, electronics, calibration residual, detector noise, or sky solely because it is correlated.

SCI-PTC-REQ-021 – Removed subspace publication. Every transformed product shall publish the complete removed subspace or template family, group, metric, rank, support, fit order, degeneracy, and gauge-equivalence class.

SCI-PTC-REQ-022 – Additive reference and null space. Every product shall publish additive-reference state, irrecoverable centering modes, complete declared null-space status, and astronomical modes known or permitted to be attenuated.

SCI-PTC-REQ-023 – Centering contract. Each centering operation shall declare axis, population, support, weights, location estimator, masks, boundaries, units, restoration rule, gauge, and resulting null space.

SCI-PTC-REQ-024 – Scaling contract. Each scaling operation shall declare axis, population, support, weights, scale estimator, nonzero/finite rule, units, inverse, gauge, and failure state; internal scaling shall be restored before ordinary output.

SCI-PTC-REQ-025 – Gauge-invariant claim. Scientific claims shall attach to the fitted correlated signal and removed subspace, not to unpinned basis sign, reciprocal basis/loading scale, component order, or rotation inside a degenerate subspace.

SCI-PTC-REQ-026 – Estimator-family identity. The resolved selected model shall name the estimator family, its exact mathematical definition, and its distinct fit and application rules; robust common mode, fixed-template regression, masked/weighted PCA/SVD, frozen-component subtraction, and diagnostic- r PCA shall not be conflated.

SCI-PTC-REQ-027 – Robust common-mode family. A common-mode family shall state the input coordinate, location estimator, aggregation weights and their unit/normalization, support, denominator rule, detector loading model, units, and robust-loss behavior; an aggregation weight shall not be labeled a fitted loading, and a median shall not be documented as an arithmetic mean.

SCI-PTC-REQ-028 – Fixed-template family. A fixed-template regression shall state template parent, unit, gauge, metric, generalized inverse, tolerance, fit support, loading support, application support, and exact treatment of rank deficiency.

SCI-PTC-REQ-029 – Masked/weighted PCA family. A PCA/SVD family shall state matrix orientation, the space containing the removed basis, every frozen quantity, every coefficient recomputed linearly at application, exact temporal-left, detector-right, two-sided, detector/time-specific, or vectorized application map, centering/scaling, missing-data model, weights/noise metric, candidate construction, component order, rank, degeneracy/tie rule, generalized inverse/tolerance, fit support, and application support. Base-v0.1 application shall project through the frozen realized subspace and metric rather than subtract one frozen numerical reconstruction.

SCI-PTC-REQ-030 – Diagnostic-only conditioned r . Conditioned- r PCA shall be available only with a separately authorized compatible parent and shall remain inert or advisory: it shall not alter calibrated- x membership, subtraction, output, coefficients, or supply an x subtraction basis.

SCI-PTC-REQ-031 – Hierarchical groups. The realized model shall identify array-wide, network/electronics, and optional local/focal-plane components and the exact detectors, coordinates, and supports in each group.

SCI-PTC-REQ-032 – Joint or sequential order. Overlapping hierarchical components shall declare joint fitting or exact sequential order. Sequential residual fitting shall be one complete ordered estimator with cumulative removed subspace, response, covariance, and parentage.

SCI-PTC-REQ-033 – Data-derived grouping. A data-derived group shall record population, metric, candidate rule, cardinality controls, stability test, held-out or adjacent-segment status, selection uncertainty, and frozen application identity.

9.4 Mode selection and source protection

SCI-PTC-REQ-034 – Finite candidate set. Mode/component selection shall evaluate a finite declared candidate set after centering, scaling, grouping, supports, and noise metric are fixed.

SCI-PTC-REQ-035 – Conjunctive admission. The selected candidate shall be the least aggressive candidate for which every required residual-contamination, astronomical-transfer, conditioning, support, stability, and QC predicate passes. Each predicate shall retain its typed value, unit, uncertainty treatment, threshold, comparison polarity, pass/fail cause, and claim.

SCI-PTC-REQ-036 – No compensating score. Failure of any required selection predicate shall not be offset by improvement in another predicate or by a scalar aggregate score.

SCI-PTC-REQ-037 – Ordering and ties. Candidate ordering, least-aggressive relation, deterministic tie handling, and boundary comparison polarity shall be declared before selection.

SCI-PTC-REQ-038 – Nonnested candidates. Nonnested candidates shall be compared by their complete removed subspace and astronomical response rather than mode count alone.

SCI-PTC-REQ-039 – Empty admission set. If no candidate passes every required predicate, the requested transformed product shall be unavailable or rejected with cause; an aggressive fallback shall not be selected silently.

SCI-PTC-REQ-040 – No universal threshold. No universal mode count, variance fraction, eigengap, or singular-value threshold shall be treated as scientific authority outside its declared model and evidence.

SCI-PTC-REQ-041 – Source-mask limit. A source mask shall protect only its declared source model and registered support and shall not be represented as proof that unmasked compact emission, extended emission, or other sky modes survive.

9.5 Post-fit detector assessment and repetition

SCI-PTC-REQ-042 – Diagnostic definition. Every residual, loading, influence, stability, source-response, non-Gaussianity, transient, or x/r diagnostic shall declare estimator, population or approved noise/signal reference, normalization, support, uncertainty, and policy role.

SCI-PTC-REQ-043 – Pathology discrimination. Post-fit assessment shall distinguish detector pathology from astronomical signal, source-model/mask failure, calibration state, focal-plane position, and expected detector-sensitivity variation.

SCI-PTC-REQ-044 – Owner-controlled thresholds. Numerical classification thresholds shall be explicit owner-controlled inputs. A missing threshold shall not be inferred from current data, code, or population behavior.

SCI-PTC-REQ-045 – Finite refinement. The fit-diagnose-classify-refit process shall be finite and shall record each support, diagnostic, class, refit decision, and emitted final state.

SCI-PTC-REQ-046 – Immutable-parent refit. Every fit-support-changing refinement shall refit one complete selected model from the same immutable admitted CAL parent and apply the final complete model once. A previously cleaned output shall not be the refit parent.

SCI-PTC-REQ-047 – Non-fit dispositions. A postfit-output rejection, coefficient-only decision, or advisory diagnostic shall not retroactively change the fitted state.

SCI-PTC-REQ-048 – Refinement stopping state. Stop state shall distinguish stable support/subspace, no-new-threshold event, maximum refinements, oscillation, insufficient support, nonconvergence, exhaustion, and error; oscillation shall not be called convergence.

SCI-PTC-REQ-049 – Internal iteration identity. Internal numerical iteration shall record statistic, norm/support, tolerance and unit, minimum/maximum iterations, comparison polarity, empty/non-finite behavior, selected iterate, and converged/exhausted/error status.

SCI-PTC-REQ-050 – PTC pass identity. A new PTC pass shall have an immutable new input parent and distinct realized identity; it shall not overwrite a previous pass.

SCI-PTC-REQ-051 – External recurrence boundary. Model subtraction/add-back, map feedback, external recurrence, restart, and recurrence parentage shall remain FRUIT or another declared recurrence owner’s authority.

9.6 Coefficients and uncertainty

SCI-PTC-REQ-052 – Coefficient taxonomy. Fitted loadings, centering/scaling parameters, diagnostic coefficients, and analysis/gridding coefficients shall be distinct typed families. Only an explicitly named analysis/gridding family may be MAP-facing.

SCI-PTC-REQ-053 – Fitted-loading semantics. A fitted loading shall declare basis/template identity, gauge, unit, group, fit support, lifecycle, and numerical use and shall prohibit weight, precision, significance, sensitivity, or independent-noise interpretation.

SCI-PTC-REQ-054 – Analysis/gridding coefficient. Each analysis/gridding family shall declare detector or detector-time index family, exact statistic/factors, their domains, unit, normalization operator, normalization domain, selected support, lifecycle, permitted consumers, numerical use, and prohibited interpretations.

SCI-PTC-REQ-055 – Coefficient re-estimation. Re-estimating any coefficient shall create a new realized product state and shall not silently modify an earlier transformed signal or parent.

SCI-PTC-REQ-056 – Empirical statistic limit. An empirical residual statistic shall declare centering, coefficients, support, effective degrees of freedom, correlation treatment, and lifecycle and shall not be labeled unbiased variance or precision without proof.

SCI-PTC-REQ-057 – Uncertainty taxonomy. Formal covariance, marginal variance, empirical scatter, calibration/systematic uncertainty, response uncertainty, selection uncertainty, and cross-coordinate covariance shall be distinct typed states.

SCI-PTC-REQ-058 – Conditional covariance. A covariance propagated through $J_{\Theta}[Y]$, the derivative or linear part of the exact frozen application map, shall be labeled conditional on the exact resolved selected model and shall identify axes, units, normalization, support, rank, approximation, and omitted correlations.

SCI-PTC-REQ-059 – Selection uncertainty. When fitted modes, grouping, rank, masks, classification, or coefficients vary, omission of the between-selection term in Equation 21 shall be declared; unknown selection uncertainty shall not be zero.

SCI-PTC-REQ-060 – Cross-observation covariance. An absent cross-observation covariance block shall mean unknown/unavailable, not independence; shared calibration, atmosphere, model, APT, or processing state shall remain possible correlation causes.

9.7 Response and transfer

SCI-PTC-REQ-061 – Admitted upstream response. A complete chain response shall begin from $K_{\text{up} \rightarrow \text{CAL}}$, the complete admitted upstream response carried by the CAL product, with explicit source domain and CAL detector-time codomain. It shall compose every applicable admitted source-to-detector relation, beam convention, coordinate and scan geometry, RTC operation, and CAL operation, or the affected complete-response claim shall be unavailable. A supplied companion already on the CAL grid shall instead enter the PTC-local operator directly.

SCI-PTC-REQ-062 – Fixed-state companion. A fixed-state companion shall be acted on by $J_{\Theta}[Y]$, the derivative or linear part of the exact frozen application map, with the same groups, coordinates, metrics, supports, masks, rank, subspace, detector classes, boundaries, and fallbacks. It shall not enter or alter learning, and $K_{\text{up} \rightarrow \text{CAL}}$ shall not be applied to a CAL-grid companion a second time.

SCI-PTC-REQ-063 – Full-procedure response. A full-procedure response shall rerun complete PTC fitting, selection, rank, classification, refinement, and application from the immutable admitted CAL parent under a declared perturbation. Its numerical finite difference shall use only the transformed-signal projection $\mathcal{P}_Z$; changed discrete or mixed state shall be retained separately through Δ_{state} .

SCI-PTC-REQ-064 – Response family. Where response depends materially on amplitude, perturbation side, location, source state, mask, rank, classification, or convergence, PTC shall publish a bounded response family or mark a stronger response unavailable.

SCI-PTC-REQ-065 – Package response boundary. A whole-chain RTC-to-CAL-to-PTC injection shall be a separately named cross-package study and shall not be relabeled PTC-only full-procedure response.

SCI-PTC-REQ-066 – Response status. Every requested response role shall use exactly one availability status among computed/published, not computed or not requested for this product, invalid, and unavailable, with cause and validity domain. Response availability shall remain distinct from product realization state.

SCI-PTC-REQ-067 – Map-center diagnostic. The optional estimated map-center point-source response shall require an exact source companion and parent domain, explicit fixed-state or full-procedure response role, complete propagated PTC companion, exact named reference MAP operator, center selector, band/mode/state binding, and uncertainty status.

SCI-PTC-REQ-068 – No response overclaim. A map-center or compact-source response shall not establish general PTC, off-center, spatially varying, extended-source, arbitrary-morphology, cross-band, cross-mode, beam-shape, or MAP authority.

9.8 Outputs, provenance, and boundaries

SCI-PTC-REQ-069 – Transformed intermediate. The in-memory transformed TOD shall be the authoritative PTC-to-MAP intermediate on the PTC-dependent route and shall not be labeled an independent sky estimator.

SCI-PTC-REQ-070 – Persisted role. A persisted PTC TOD shall declare either diagnostic artifact or requested derived analysis product; persistence shall not grant archival, map, beam, significance, or production authority.

SCI-PTC-REQ-071 – Material provenance. All material state affecting a declared consumer shall be linked to immutable parent, requested, effective-policy, observation-resolved, learned/fitted evidence, resolved selected model, applied/realized, and published states, plus supports, groups, modes, rank, coefficients, classification, response, covariance, randomness, and output identity.

SCI-PTC-REQ-072 – Atomic required outputs. A required product shall be complete only after all required components and links are durable. Partial publication or required-output failure shall propagate and shall not be labeled complete.

SCI-PTC-REQ-073 – Typed fallback. Every fallback shall declare cause, stage, changed estimator/state, response effect, covariance effect, and affected availability; an unavailable support shall not become valid zero output.

SCI-PTC-REQ-074 – Insufficient support. Empty, non-finite, zero-denominator, rank-deficient-beyond-policy, or otherwise insufficient support shall yield a declared unavailable/rejected state or an explicitly authorized typed fallback.

SCI-PTC-REQ-075 – Random state. Randomness capable of changing science output shall record generator family/version, seed, stream partition, draw identity, and input identity; repeated declared state shall be reproducible.

SCI-PTC-REQ-076 – Disabled PTC route. When PTC is disabled, named PTC products on the PTC-dependent route shall have product realization state **disabled**; no PTC centering, subtraction, coefficient, transformed product, or PTC-route MAP product shall be realized. Response availability shall be recorded on its separate axis and shall not be encoded as product nonexistence.

SCI-PTC-REQ-077 – Direct CAL-to-MAP separation. No requirement in this contract shall prohibit or authorize a separately governed direct CAL-to-MAP route.

SCI-PTC-REQ-078 – Excluded signal roles. Base v0.1 shall not authorize raw- $\Delta f/f$ Beammap PTC, raw- r numerical science, r -derived subtraction from x , unconstrained joint x/r PCA, cross-array fitting, or polarimetry.

SCI-PTC-REQ-079 – Ownership preservation. Consuming another package's product shall not transfer ownership of RTC temporal conditioning, CAL calibration/atmosphere, AST/ALIGN coordinates, VAL eligibility, MAP estimation, NOI uncertainty, BEAM interpretation, FLT filtering, or FRUIT recurrence.

SCI-PTC-REQ-080 – Evidence-layer separation. The contract shall distinguish algebraic contract correctness, implementation conformity, representation/response fidelity, observational performance, and production readiness. This document shall claim only the first.

9.9 Explicit scientific-review scope-completion obligations

SCI-PTC-REQ-081 – External source/residual parent. Any source model, mask, residual, or prior-pass parent consumed for protection or fitting shall declare owner, product and recurrence identity, unit, coordinate frame, registration, validity, response, use-specific support, boundary rule, and relation to the immutable CAL parent. Missing required identity shall make the affected protected or fitted role unavailable.

SCI-PTC-REQ-082 – Shifted/null surrogate. A shifted or null surrogate shall transform signal and its associated validity/eligibility, support, boundaries, randomness, and parent identity together under one declared rule. Insufficient transformed support shall be unavailable or rejected and shall not become a valid numerical-zero surrogate.

SCI-PTC-REQ-083 – Exact frozen application map. Every realized estimator shall publish $\mathcal{A}_\Theta$ and $J_\Theta[Y]$ with exact input/output domains, left/right/vectorized action, frozen quantities, recomputed coefficients, metric, supports, affine reference, and failure behavior. Equality to $Y - \hat{U}$ on the fitted parent shall not substitute for this application contract.

SCI-PTC-REQ-084 – Candidate evidence retention. The learned/fitted evidence state shall retain every evaluated candidate, its fitted state, diagnostics, every predicate value and pass/fail cause, ordering and tie result, selected or rejected disposition, and the exact immutable application map of the selected model.

SCI-PTC-REQ-085 – Typed astronomical-transfer predicate. Every astronomical-transfer predicate shall identify source class and companion, conditional or full-procedure response type, amplitude and location domain, metric, tolerance, mask/support, upstream response identity, response-uncertainty treatment, and the exact claim tested.

SCI-PTC-REQ-086 – Expectation and bias statement. Each estimator family and product role shall state its conditional expectation or bias claim for astronomical signal, fitted nuisance, remaining detector noise, calibration uncertainty, and data-dependent selection, or mark each unsupported claim unavailable. No estimator shall be labeled unbiased by omission.

SCI-PTC-REQ-087 – Response-domain chain. Every response product shall distinguish the PTC-local operator $J_\Theta[Y]$, complete chain response $J_\Theta[Y] \circ K_{\text{up} \rightarrow \text{CAL}}$, and realized propagated companion. Source-domain and CAL-grid parents shall have explicit domains, the response carried by the CAL product shall not be confused with only the CAL-owned multiplier, and each upstream operator shall be applied exactly once.

SCI-PTC-REQ-088 – Independent state axes. Product realization, response availability, covariance availability, and validation/evidence disposition shall be separate typed axes. A disabled or not-requested product shall not be encoded as an invalid response, and an unavailable response shall not imply that an otherwise supported signal product does not exist.

SCI-PTC-REQ-089 – Fit-excluded application availability. A fit-excluded occurrence shall be application-available only when the resolved frozen operator defines its value from admitted fitted state and support and every required basis coefficient or coefficient-recomputation input, neighboring support, detector binding, metric term, coordinate transform, and boundary state is available. Otherwise application at that occurrence shall be unavailable; it shall not be implicitly interpolated, zero-filled, reconstructed, or extended from the fit population.

10 Falsifiable edge predictions

These are scientific oracles for later conformance and validation work. They are not claims that an implementation has passed.

SCI-PTC-PRED-001 – Disabled route identity. With PTC disabled, each PTC signal, fitted-state, coefficient, and PTC-route MAP product has product realization state `disabled`; any requested upstream terminal product remains distinct, and response availability remains a separate typed axis.

SCI-PTC-PRED-002 – Same-unit response loss. A detector-common astronomical perturbation can retain mJy-per-fixed-nominal-beam units while being attenuated or annihilated; unit equality alone will not predict recovered amplitude.

SCI-PTC-PRED-003 – Correlation non-identification. Two data sets with the same detector covariance but different physical mixtures admit the same fitted correlated subspace; the fit alone cannot distinguish atmosphere from sky or electronics.

SCI-PTC-PRED-004 – Gauge invariance. Negating a basis/loading pair, reciprocally rescaling it, or rotating a degenerate basis leaves the fitted correlated signal and transformed output unchanged within declared numerical tolerance.

SCI-PTC-PRED-005 – Centering null mode. Changing detector-time data by a mode removed by the declared centering operator does not change the centered fit coordinate; that mode appears in the published additive-reference/null-space state.

SCI-PTC-PRED-006 – Scaling restoration. For an invertible internal scaling multiplied by a finite nonzero constant and consistently inverted, ordinary transformed output in the physical unit is unchanged within tolerance.

SCI-PTC-PRED-007 – Zero-fill is not exclusion. Replacing one excluded nonzero sample by zero before ordinary PCA changes means/covariance and can change the selected subspace; a correctly masked/weighted missing-data estimator gives zero fit influence instead.

SCI-PTC-PRED-008 – Fit-excluded apply-allowed. An occurrence excluded from fitting but admitted for application is acted on by the exact frozen application map without affecting learned or selected state; toggling only its output support does not refit.

SCI-PTC-PRED-009 – Support-stage independence. Changing coefficient-only support changes the affected coefficient product but not the fitted basis or transformed signal; changing fit support invalidates or refits the fitted state.

SCI-PTC-PRED-010 – Influence through fitting. A single occurrence used to fit a common mode can influence subtractions at many other occurrences even where its direct local derivative is zero; those outputs carry transitive fit influence.

SCI-PTC-PRED-011 – Source-mask boundary. Moving a sample across a declared mask boundary changes fit/application behavior only according to the serialized polarity and boundary rule; unmasked sky remains eligible to project into the removed subspace.

SCI-PTC-PRED-012 – Within-array isolation. Changing data in another array does not change a base-v0.1 fitted subspace, rank, subtraction, or coefficient for the selected array.

SCI-PTC-PRED-013 – Hierarchy order materiality. For overlapping nonorthogonal array, network, and local components, sequential orders can produce different cumulative subspaces; the declared order reproduces the realized result and cannot be omitted from state.

SCI-PTC-PRED-014 – Common-mode denominator failure. A group/time with zero, empty, or non-finite admitted denominator yields unavailable/rejected state or a typed fallback, never a silently valid zero common mode.

SCI-PTC-PRED-015 – Fixed-template orthogonality. For a frozen fixed-template metric, a perturbation exactly orthogonal to the removed subspace is preserved by the conditional operator; learned-template motion is a separate full-procedure effect.

SCI-PTC-PRED-016 – Rank-deficient gauge. Duplicate or zero columns produce a rank-deficient fit whose generalized inverse, tolerance, selected subspace, and deterministic tie rule reproduce the emitted subtraction.

SCI-PTC-PRED-017 – Least-aggressive selection. For nested candidates where the first two fail at least one required predicate and the third passes all, the third is selected even if an earlier candidate has a better scalar score.

SCI-PTC-PRED-018 – No compensating metric. A candidate with excellent residual suppression but failed astronomical-transfer predicate is rejected; residual improvement cannot compensate for sky-response failure.

SCI-PTC-PRED-019 – Deterministic tie. Two candidates equal under every required predicate and the declared ordering resolve identically across repeated execution using the registered tie rule.

SCI-PTC-PRED-020 – Empty candidate set. If no candidate passes all required predicates, no transformed product claiming admitted mode selection is produced.

SCI-PTC-PRED-021 – Immutable-parent refinement. A support-changing second refinement fit equals a fresh fit of the complete selected model on the immutable CAL parent and differs, in general, from fitting the already cleaned first output.

SCI-PTC-PRED-022 – Output-only classification. Changing one detector from retained to postfit-output-reject without changing fit support preserves the fitted subspace and other detectors' transformed values.

SCI-PTC-PRED-023 – Refinement oscillation. Alternating membership of a high-leverage detector reaches the declared oscillation/exhaustion state and is never labeled converged.

SCI-PTC-PRED-024 – Diagnostic population dependence. The same residual statistic can be ordinary under one declared population/noise reference and anomalous under another; the policy result changes only with an explicitly changed reference state.

SCI-PTC-PRED-025 – Diagnostic-only r inertia. Changing an admitted conditioned- r diagnostic while holding calibrated x and all x policy fixed can change only r diagnostic outputs, not x fit membership, subtraction, output, or coefficients.

SCI-PTC-PRED-026 – Raw- r incompatibility. If conditioned- r operator, unit, response, optical-leakage state, or exact x -grid relation is missing, the r diagnostic branch is unavailable while the admitted x branch may remain defined.

SCI-PTC-PRED-027 – Coefficient gauge. Reciprocal basis/loading rescaling changes fitted loading values but leaves the modeled subtraction unchanged; an analysis/gridding coefficient does not transform merely because a loading gauge changes.

SCI-PTC-PRED-028 – Coefficient normalization. Multiplying all raw gridding coefficients by a common nonzero factor can leave a normalized weighted estimate unchanged while changing the denominator, proving that the denominator is not automatically precision.

SCI-PTC-PRED-029 – Conditional covariance. For a fixed analytic $J_{\Theta}[Y]$ and declared C_Y , the conditional covariance equals $J_{\Theta}[Y]C_YJ_{\Theta}[Y]^T$ including off-diagonal detector/time terms.

SCI-PTC-PRED-030 – Selection variance term. Across perturbations that change rank or support, total output variance exceeds or differs from the mean conditional covariance whenever between-selection means vary.

SCI-PTC-PRED-031 – Fixed companion noninterference. Adding a response companion through the fixed-state path leaves fitted modes, rank, masks, detector classes, and transformed science data unchanged.

SCI-PTC-PRED-032 – Full-procedure nonlinearity. An injected source that crosses a rank, mask, clipping, classification, or convergence boundary produces amplitude- or side-dependent finite response rather than one global linear kernel.

SCI-PTC-PRED-033 – Package response boundary. Rerunning PTC from a fixed CAL parent can characterize PTC-full-procedure response but cannot expose a changed RTC or CAL learned state; that requires a separately named whole-chain study.

SCI-PTC-PRED-034 – Map-center factor availability. Removing the exact source companion or named reference MAP operator makes the map-center diagnostic unavailable while leaving a supported sample-domain PTC response potentially available.

SCI-PTC-PRED-035 – Compact versus extended response. A compact source and detector-common extended mode with equal nominal peak amplitude can have different PTC responses; the compact map-center diagnostic cannot predict the extended response.

SCI-PTC-PRED-036 – Random-state reproducibility. Repeating a simulation or randomized estimator with identical generator/version, seed, stream, and input identity reproduces the declared draws and realized result; changing one stream changes only its assigned draws.

SCI-PTC-PRED-037 – Required-output atomicity. Failure to publish one required sibling or link leaves the product explicitly incomplete; a remaining readable file cannot be labeled complete.

SCI-PTC-PRED-038 – Evidence-layer separation. Successful algebraic and document-coverage checks do not change implementation conformity, representation fidelity, observational performance, or production-readiness status from unassessed/not performed.

SCI-PTC-PRED-039 – Frozen component versus projection. For a perturbation with nonzero projection onto the frozen removed subspace, frozen-component subtraction has identity derivative while frozen-subspace application removes that projected component; the two responses differ even though both reproduce $Y - \hat{U}$ on the fitted parent.

SCI-PTC-PRED-040 – Temporal versus detector action. For a nonsymmetric response matrix H , temporal-left projection $\mathcal{L}_{\Theta}^t H$ and detector-right projection $H \mathcal{L}_{\Theta}^d$ generally differ; matrix orientation and rank alone cannot select the propagated companion.

SCI-PTC-PRED-041 – No double upstream response. Propagating a CAL-grid companion through $J_{\Theta}[Y]$ gives the same result as propagating its source-domain parent through $J_{\Theta}[Y] \circ K_{\text{up} \rightarrow \text{CAL}}$ exactly once; applying $K_{\text{up} \rightarrow \text{CAL}}$ again changes the domain or value and is rejected.

SCI-PTC-PRED-042 – Typed procedure state change. A perturbation that changes selected rank yields a valid transformed-signal finite difference and a separate typed rank change; attempting to subtract the complete procedure tuples is undefined and cannot produce the response.

SCI-PTC-PRED-043 – Candidate-evidence reconstruction. Given the retained candidate order, fitted states, predicate values, causes, and tie rule, the selected or rejected disposition is reproduced exactly; removing one failed predicate record makes the selection audit incomplete.

SCI-PTC-PRED-044 – External-parent fail-closed. Removing the recurrence identity or coordinate registration from an otherwise numerical source/residual parent makes the affected protected or fitted role unavailable while leaving independent PTC roles unchanged.

SCI-PTC-PRED-045 – Surrogate support transformation. Shifting surrogate signal without shifting its associated validity/eligibility changes admitted support and fails the surrogate contract; shifting both together reproduces the declared transformed membership, while insufficient support is rejected rather than zero-filled.

SCI-PTC-PRED-046 – Astronomical-predicate typing. Two candidates with the same scalar transfer value but different source class, response type, amplitude/location domain, or uncertainty treatment do not satisfy the same astronomical-transfer predicate and cannot be substituted.

SCI-PTC-PRED-047 – Unavailable bias claim. When an estimator family supplies no expectation/bias statement for data-dependent selection, the bias claim remains unavailable even if one synthetic realization has zero measured residual mean.

SCI-PTC-PRED-048 – Independent product and response states. An otherwise realized transformed signal can coexist with unavailable complete response, while a disabled PTC product has no realized signal even if an upstream CAL response exists; neither state can be inferred from the other.

SCI-PTC-PRED-049 – Undefined fit-excluded extension. For a fit-excluded occurrence missing any basis coefficient or coefficient-recomputation input, neighboring support, detector binding, metric term, coordinate transform, or boundary state required by the resolved operator, application is unavailable. Filling, interpolating, or reconstructing that occurrence changes the declared operator and cannot reproduce the authorized fit-excluded/apply-allowed result.